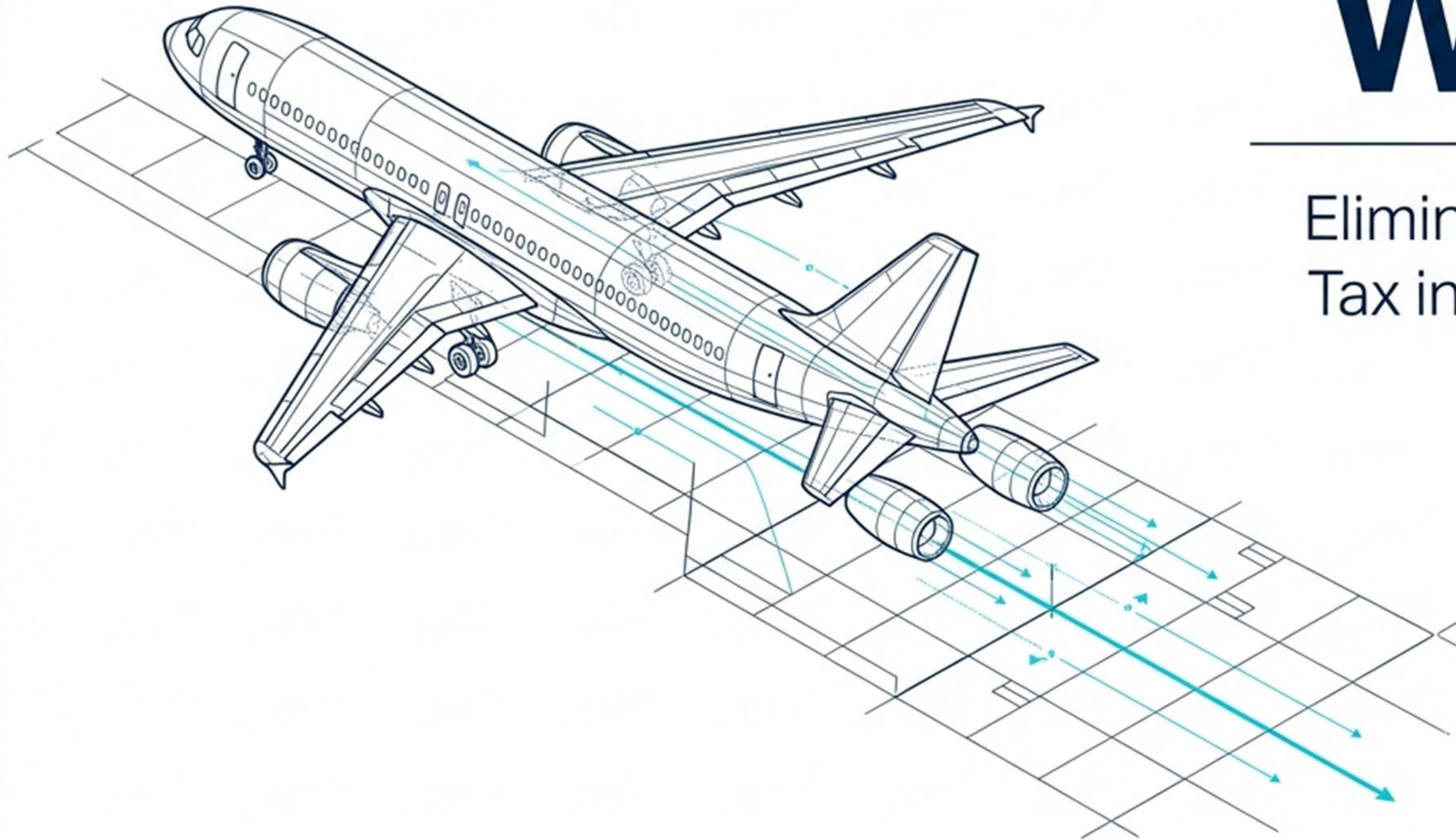


DEFINITE WEIGHT

Eliminating the Uncertainty
Tax in Commercial Aviation



Strategic Briefing & Executive Summary // Confidential

Executive Summary of the Definite Weight Business Case



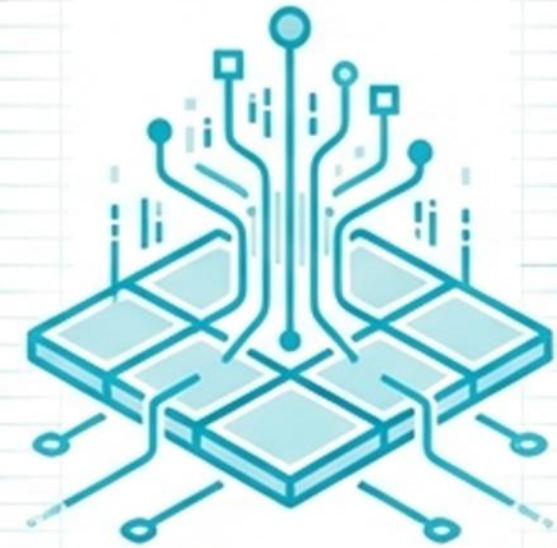
The Flaw

For 50+ years, aviation has **guessed payload mass** using **Standard Average Weights** (e.g., 84 kg/passenger). This mandates loading **thousands of pounds of unnecessary contingency fuel** to transport non-existent ghost weight.



The Cost

On a 100-aircraft fleet, this flaw actively destroys margins, costing **\$3.7M** in **wasted fuel**, **\$470K** in **CORSIA/EU ETS penalties**, and **>\$100M** in **forfeited cargo revenue** annually.



The Solution

Definite Weight is a **zero-install, SaaS-driven solution** capturing **exact mass** via invisible **biometric gate mats**. It delivers **100% precision fueling, zero passenger friction, and immediate margin expansion**.

The Status Quo Forces Airlines to Burn Fuel Carrying Ghost Weight

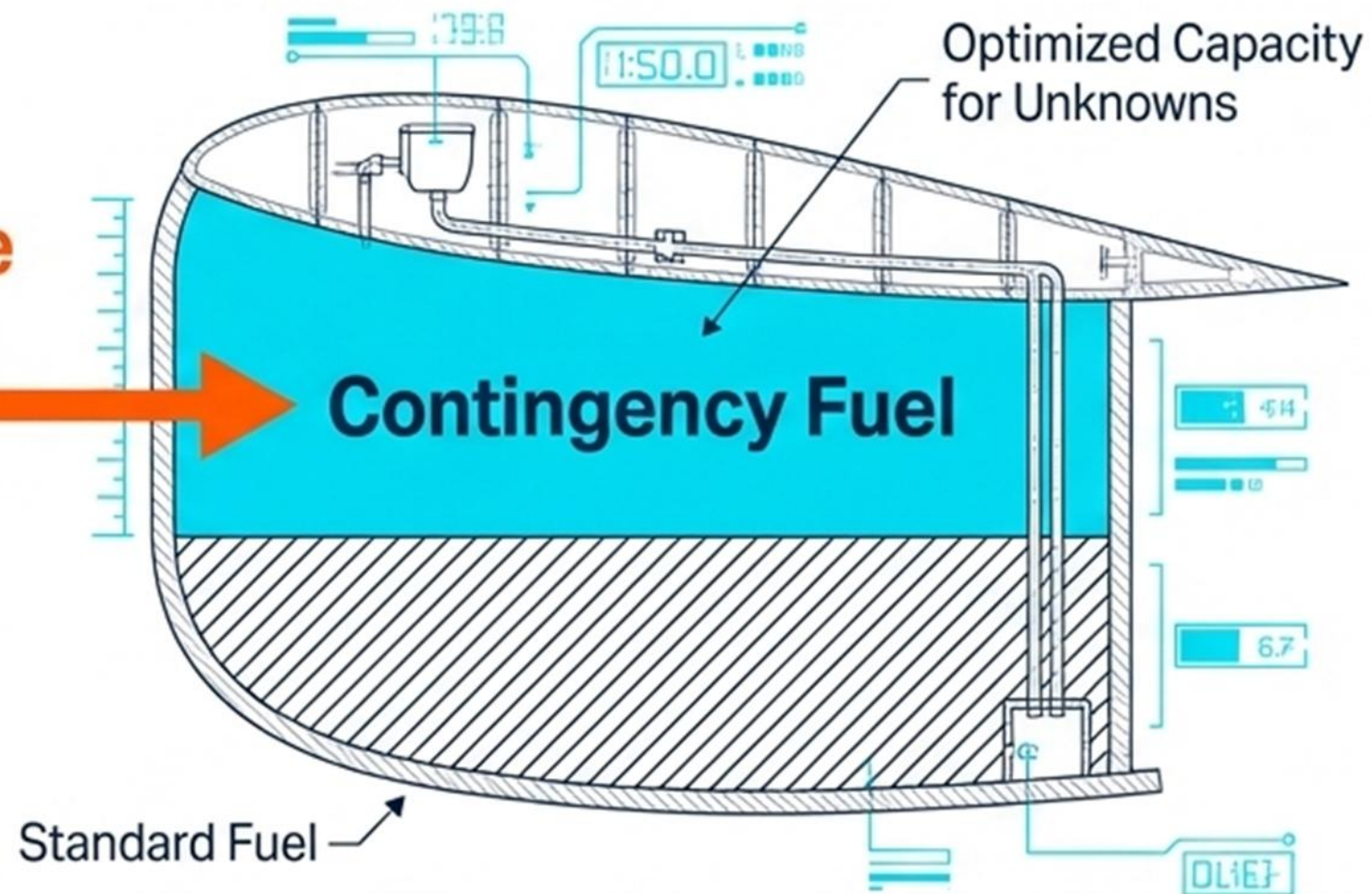
The Standard Average

84 kg

Aviation payload calculations rely on generalized standard averages, making true individual passenger weight an **unknown variable**.

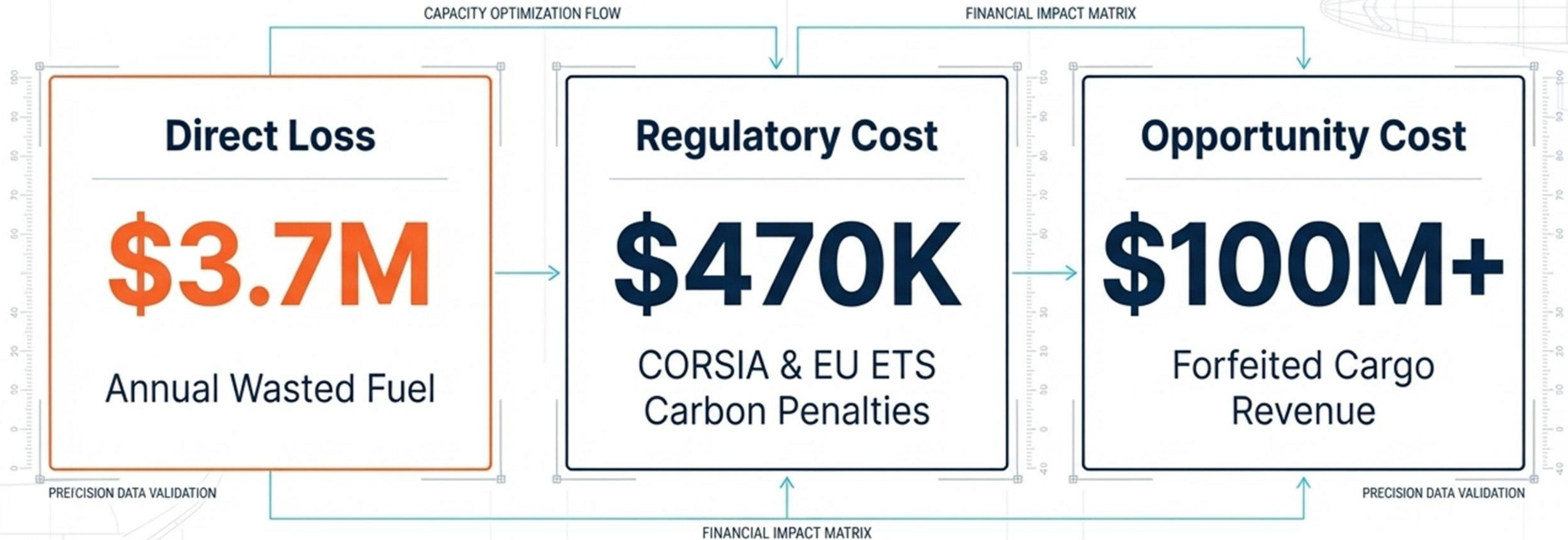
Forces Extreme Over-Fueling

The Reality



To compensate for data blindness, safety regulations mandate **extreme conservatism**. The airline pays a daily **Uncertainty Tax** in the form of fuel burn to fly an imagined payload.

Quantifying the Uncertainty Tax on a Standard 100-Aircraft Fleet



This is not a micro-optimization; the Uncertainty Tax represents a massive, systemic P&L vulnerability operating at the core of current flight planning.

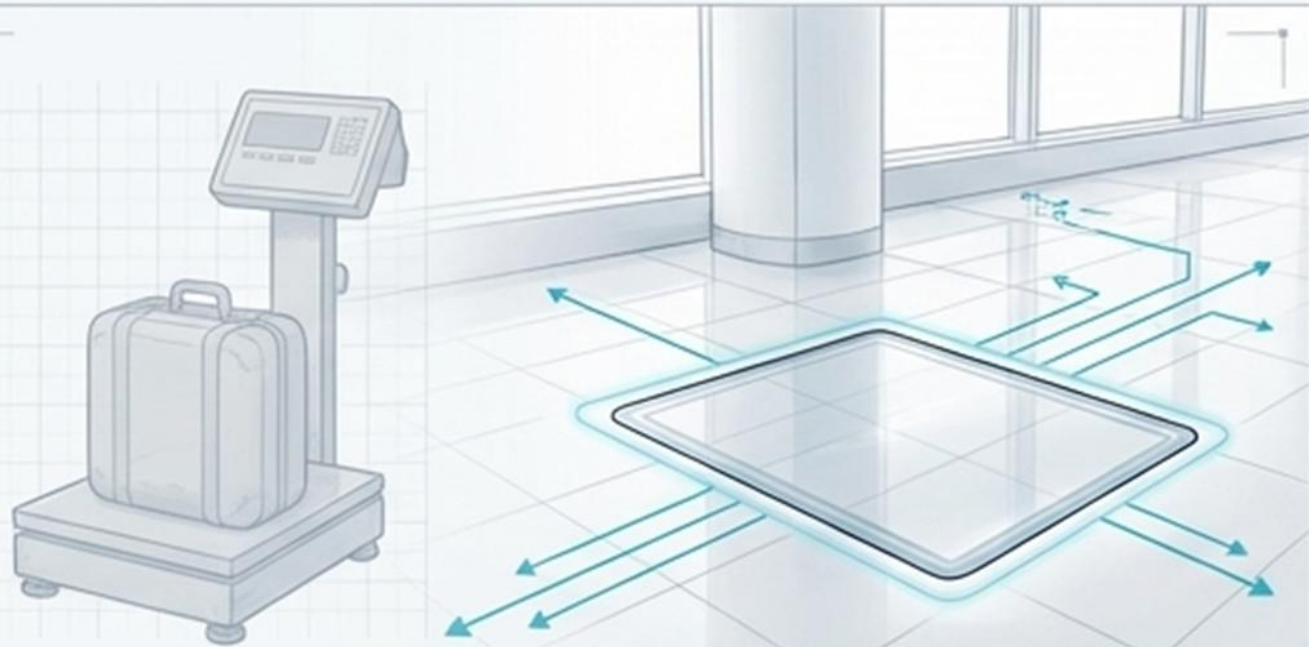
Previous payload accuracy initiatives failed by misunderstanding passenger psychology, terminal physics, or software integration

The Payload Accuracy Landscape

Methodology	Passenger Friction	Regulatory Accuracy (Physics)	Software Ecosystem & CapEx
Standard Average (Status Quo)	✓ Zero Friction	! Abysmal (Massive Fuel Waste)	✓ Zero CapEx
Medical Scales at Gate	! Critical Friction (PR Nightmare)	✓ High Accuracy	! Manual Data Silos
Jet Bridge Sensors	✓ Zero Friction	! Failed Physics (Bouncing Tubes)	! Extreme CapEx
Definite Weight (Smart Mats)	✓ Zero Friction	✓ Perfect Accuracy (Solid Concrete)	✓ Zero CapEx API Integration

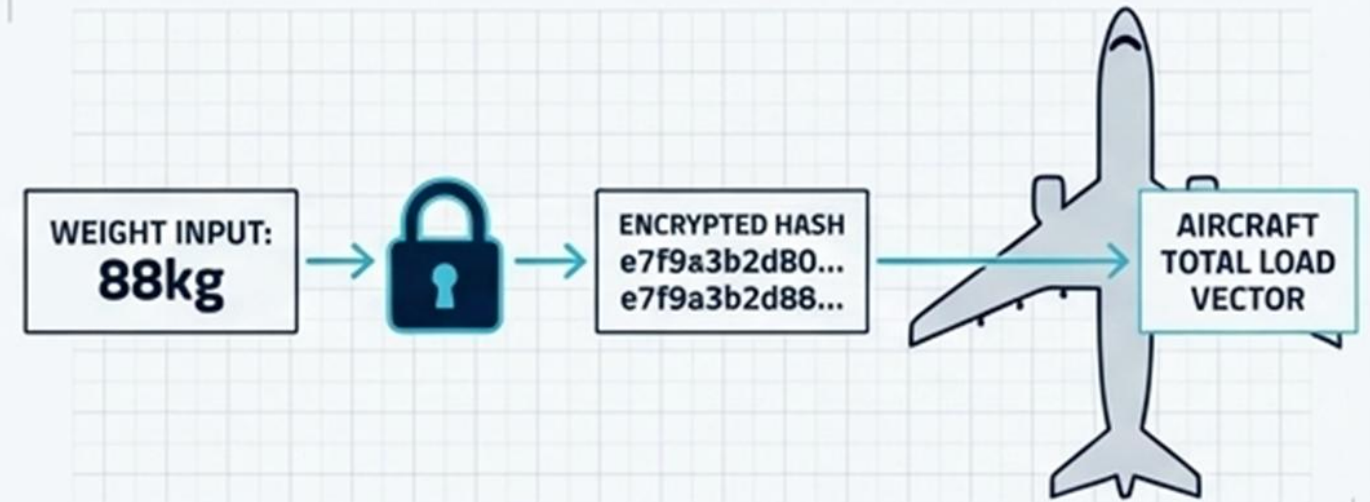
Deploying a Software Solution Through Invisible Hardware

Data-Fusion SaaS Paradigm



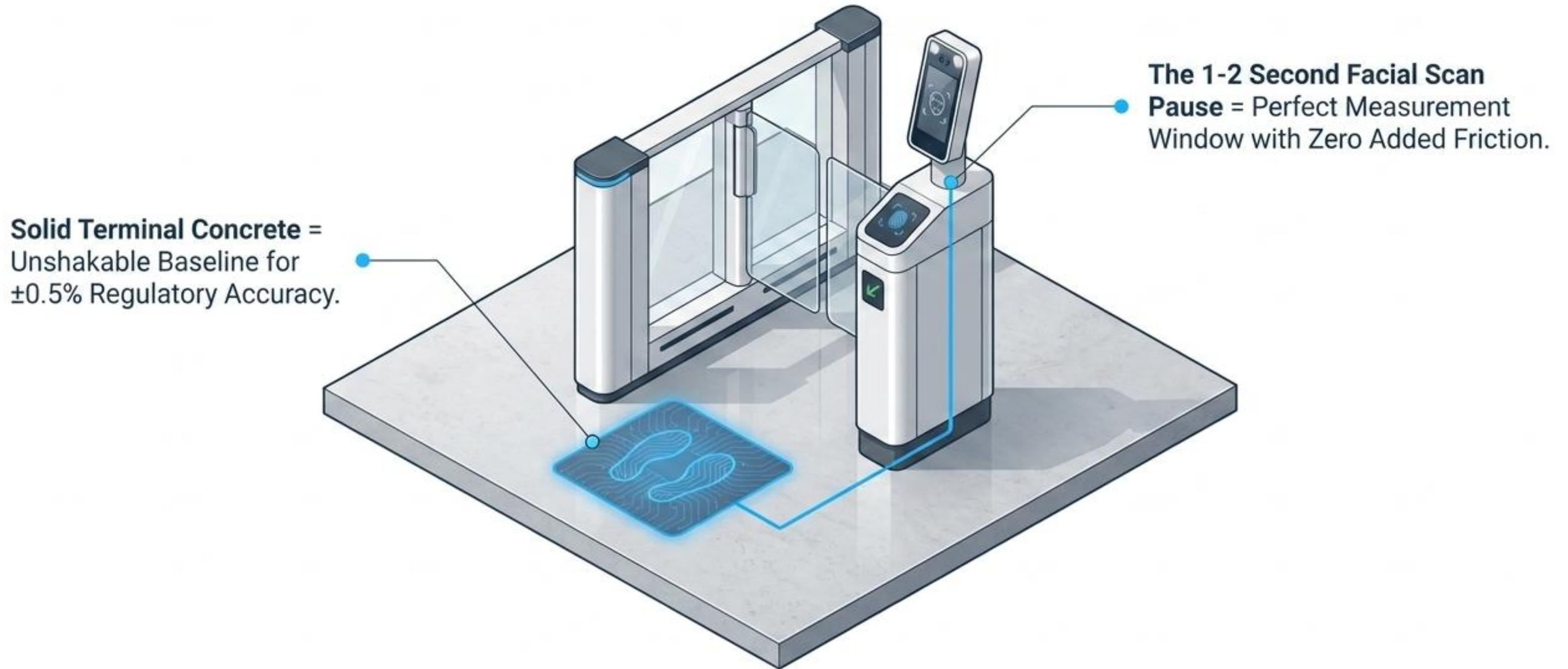
Definite Weight is not a hardware vendor. We are a data-fusion software company utilizing invisible Weigh-In-Motion (WIM) Smart Mats to capture mass without footprint.

Strict Differential Privacy



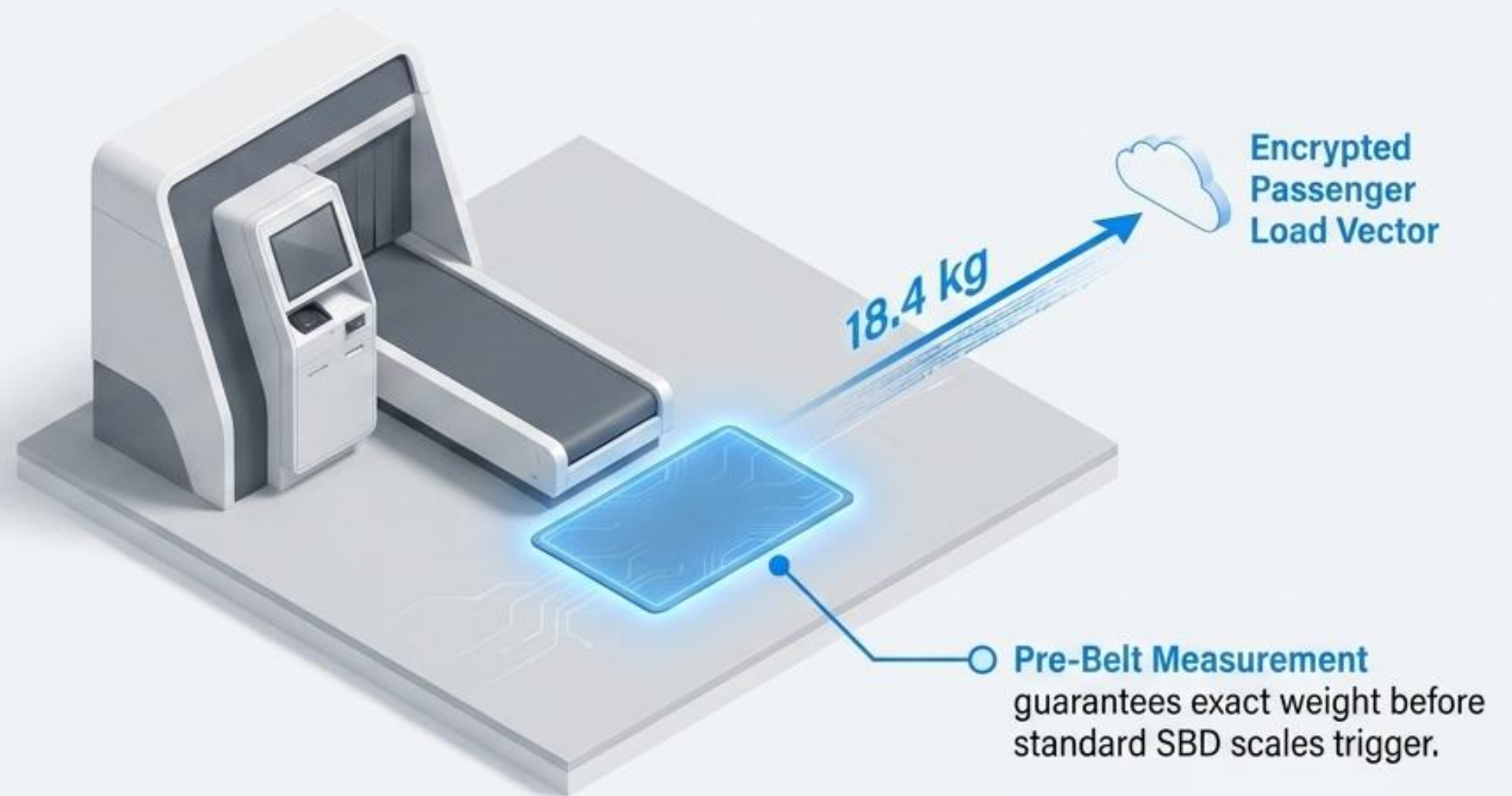
No screens display the weight. No data is tied to a passenger's name. The system captures mass, encrypts it, and adds it anonymously to the aircraft's total Load Vector. The passenger is entirely unaware.

Ultra-thin Smart Mats integrate directly into Biometric E-Gates, capturing passenger mass during the natural boarding pause



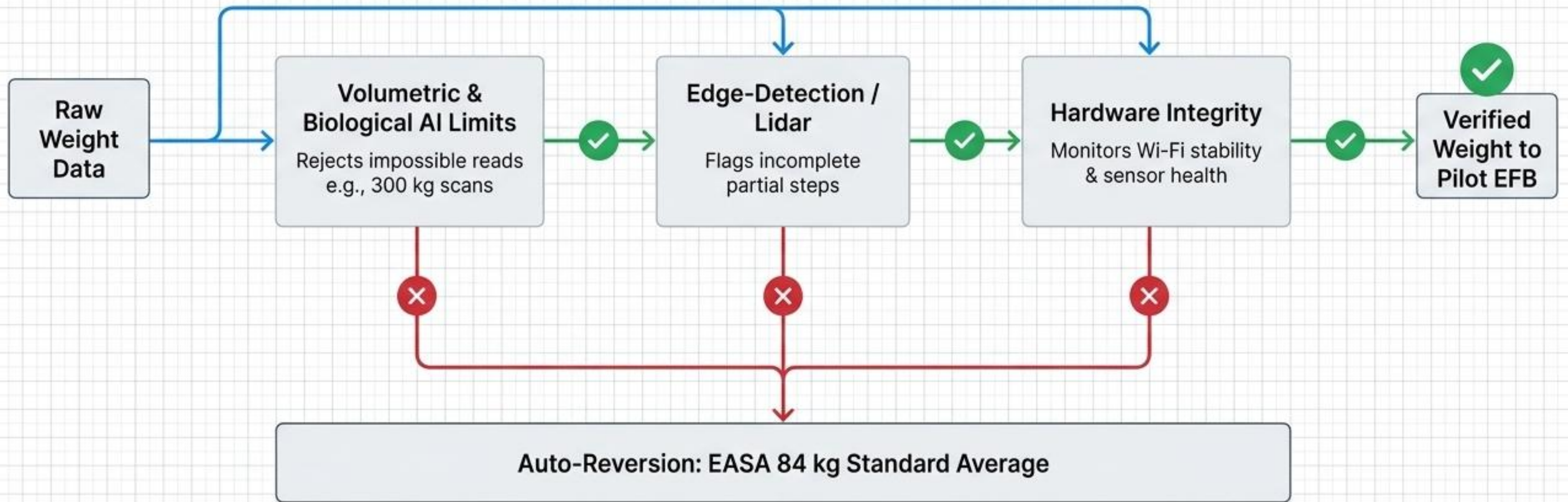
Deploying Weigh-In-Motion mats at Self-Bag Drops captures exact luggage mass before entering the terminal belt system

Capturing exact weight at the SBD eliminates standard average assumptions for heavy vs. light bags, creating a complete, end-to-end verified payload profile long before the aircraft doors close.



The API utilizes rigorous SITA-Grade sanity checks to ensure a flawed data point never reaches the flight deck

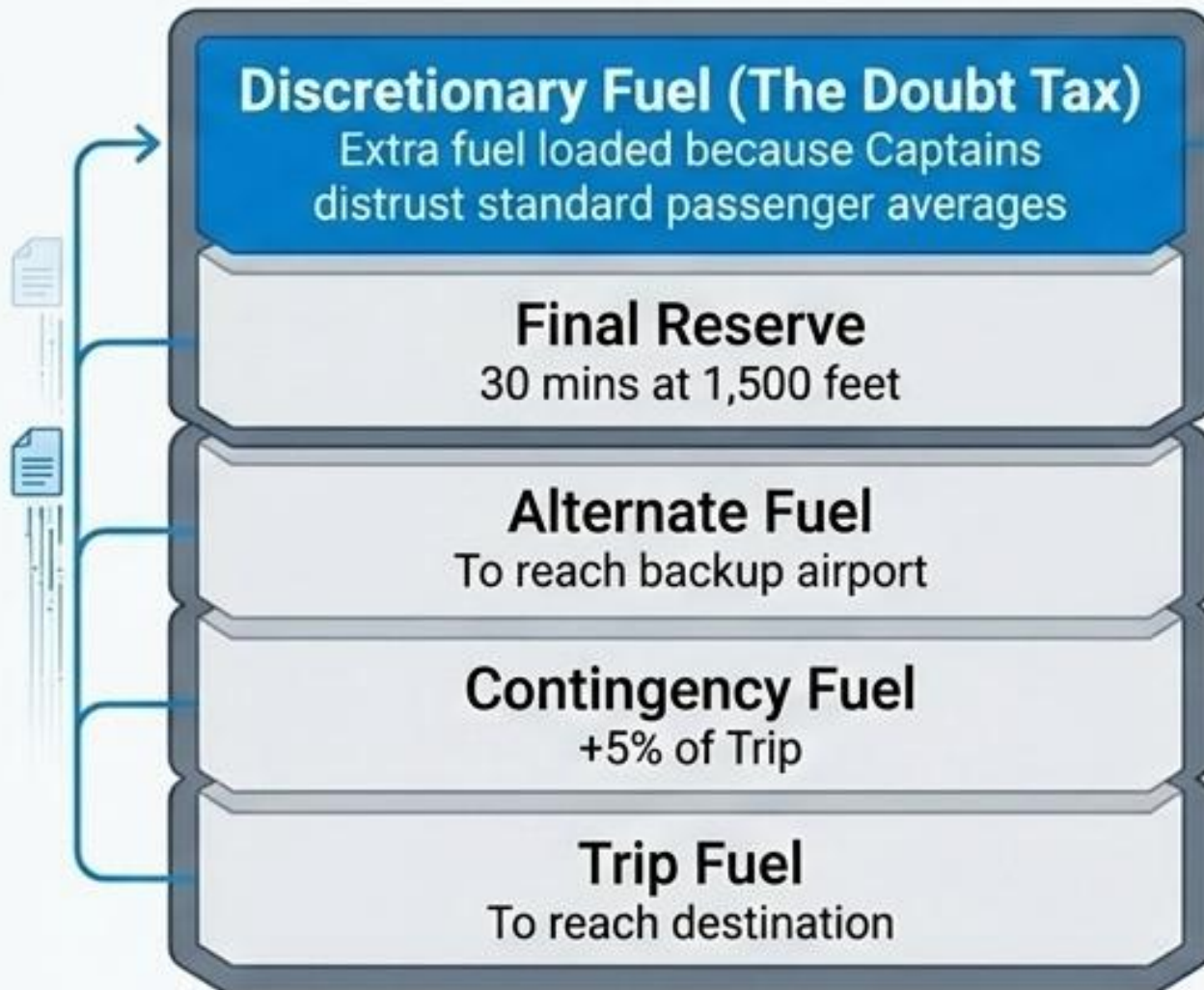
"Data Gatekeeper" Flowchart



The system is designed to **fail safe, never live**. Any anomaly instantly invalidates the sensor and reverts to the legal default.

EASA regulations mandate compounding fuel buckets, heavily penalizing estimated ghost weight

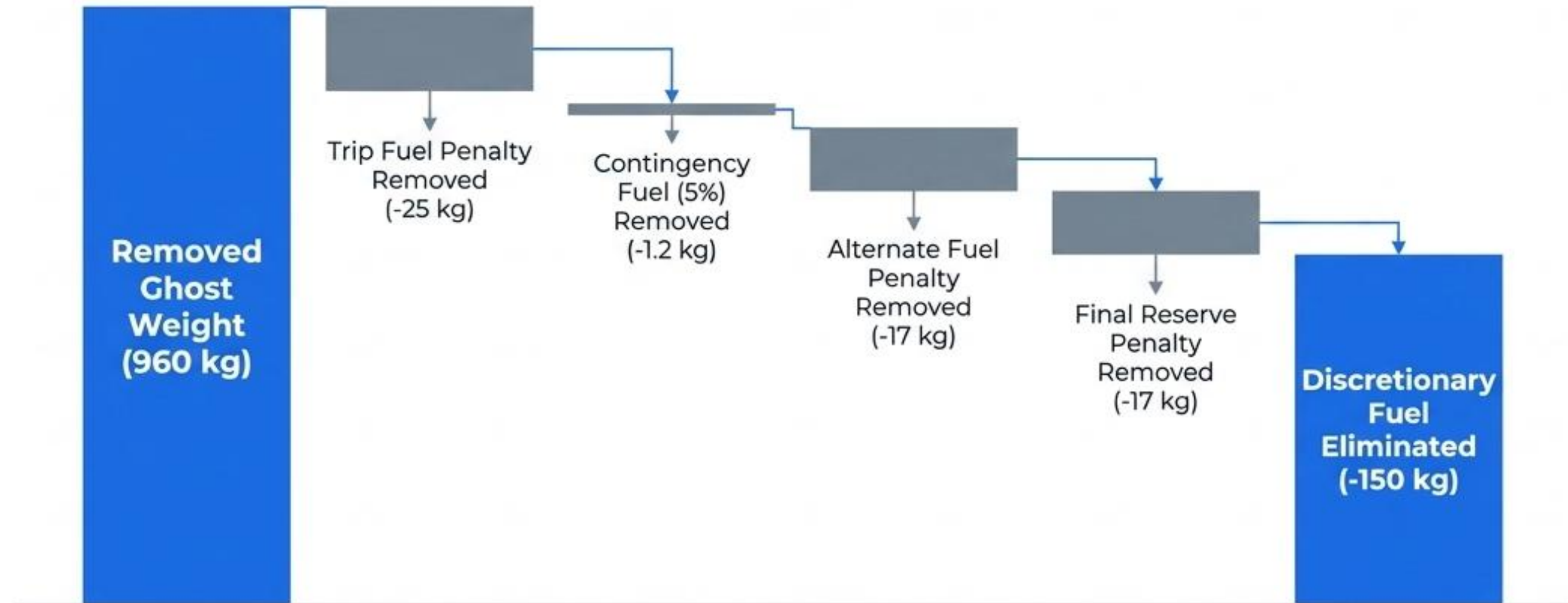
EASA Rule CAT.OP.MPA.150
Fuel Buckets (45 min flight)



The Cost of Weight

Because jet fuel is heavy, an Airbus A320neo burns ~**3.5%** of its **added weight** per hour just to stay airborne. Carrying heavy fuel buckets forces engines to **burn more fuel**.

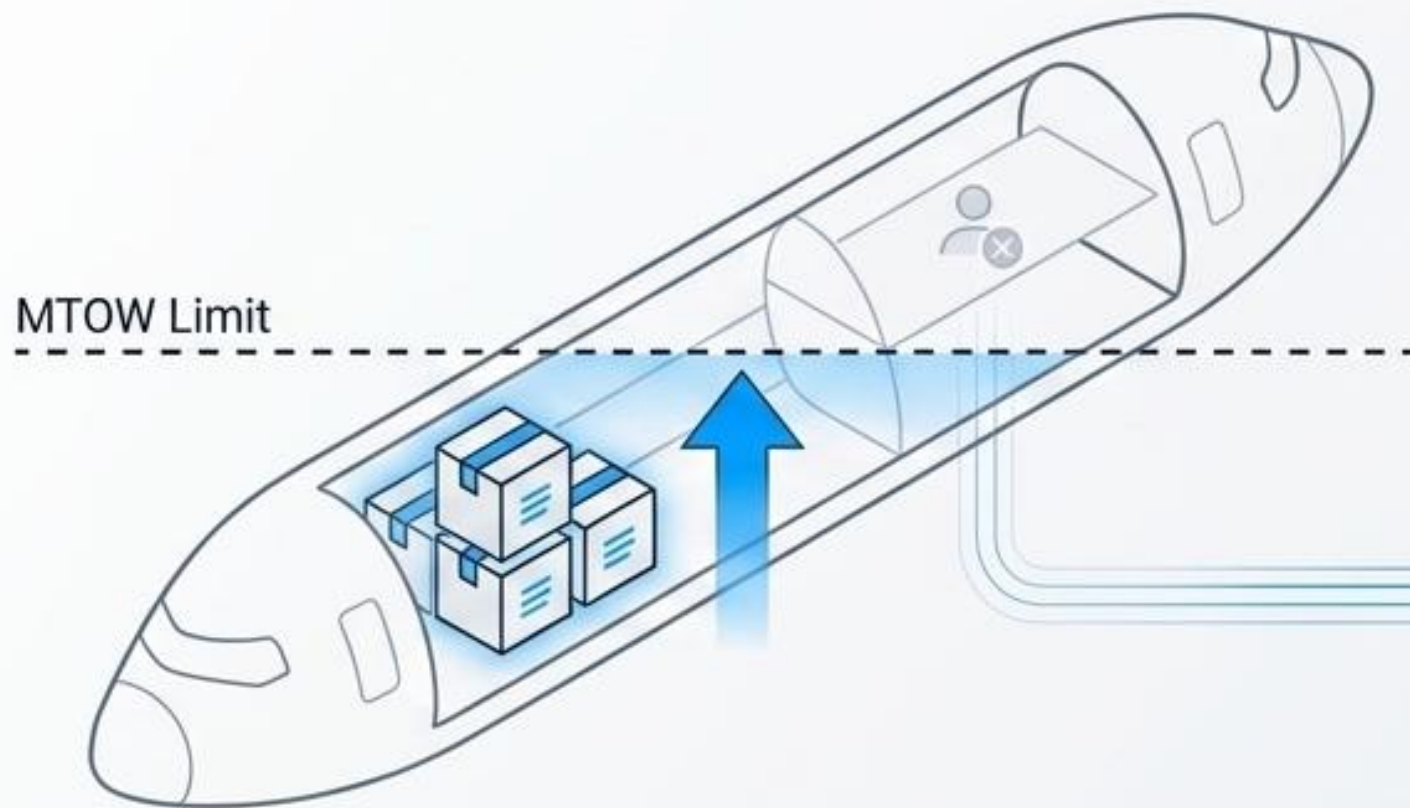
The MUC-FRA Fuel Cascade: Removing 960 kg of ghost payload triggers a mathematical cascade of savings



Net Result: Removing 960 kg of fake payload saves ~210 kg of actual, burnable fuel on a single 45-minute flight.

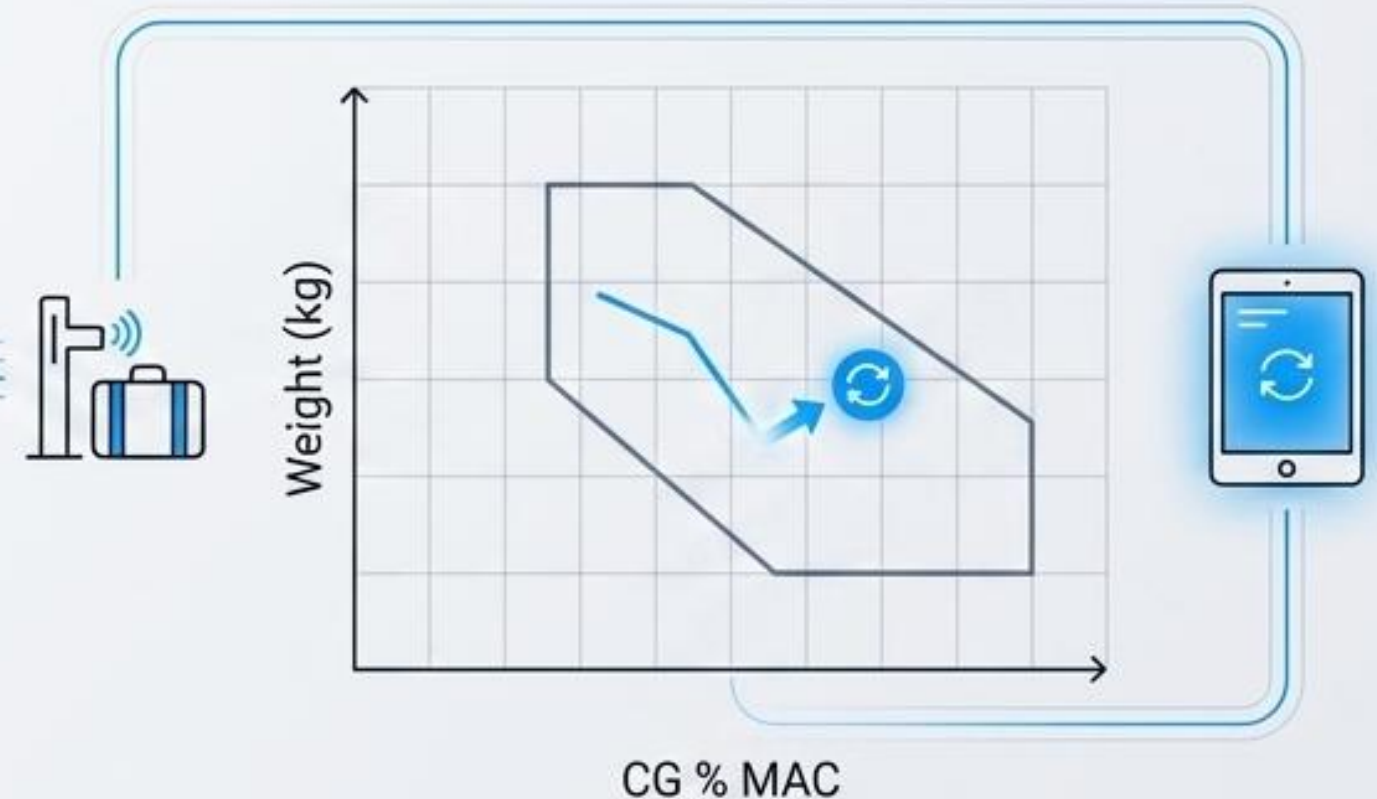
Precision payload expands operating margins by unlocking constrained cargo capacity and enhancing operational resilience

Unlocking the Cargo Bottleneck



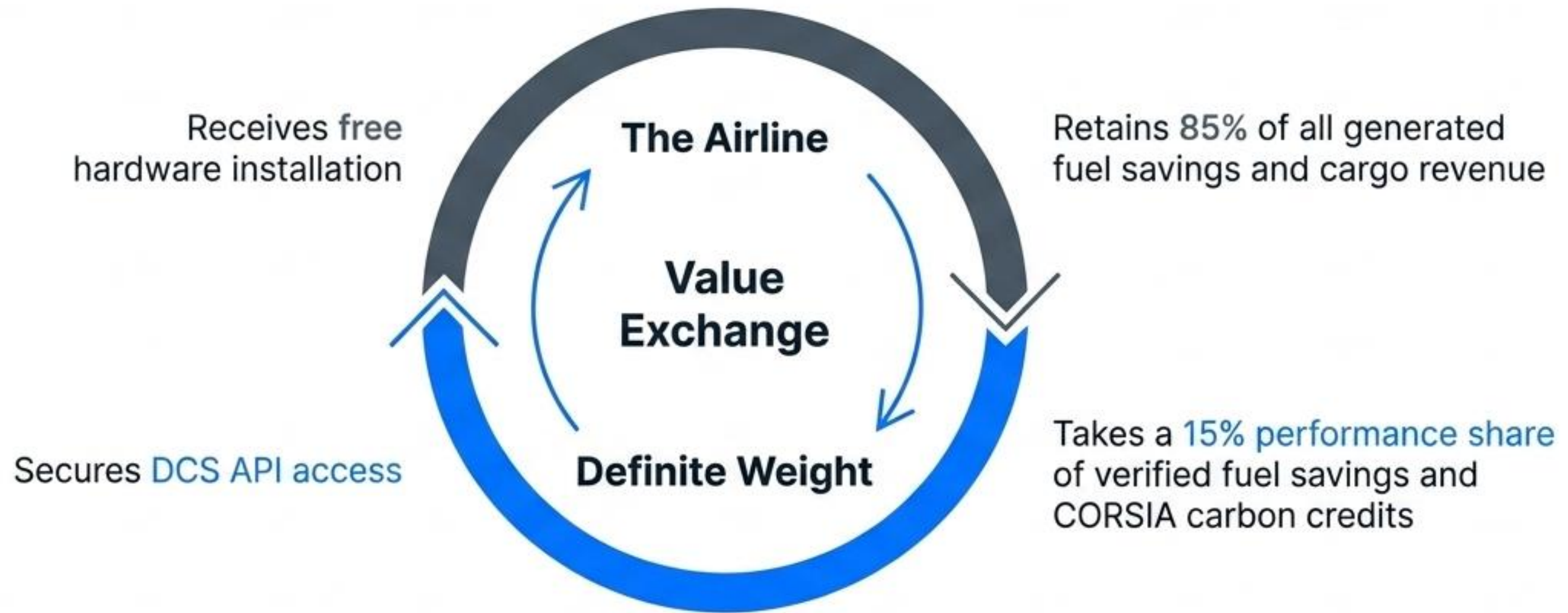
Creates room for 2 to 3 additional high-yield cargo pallets under MTOW limits by removing artificial passenger weight inflation.

Operational Resilience



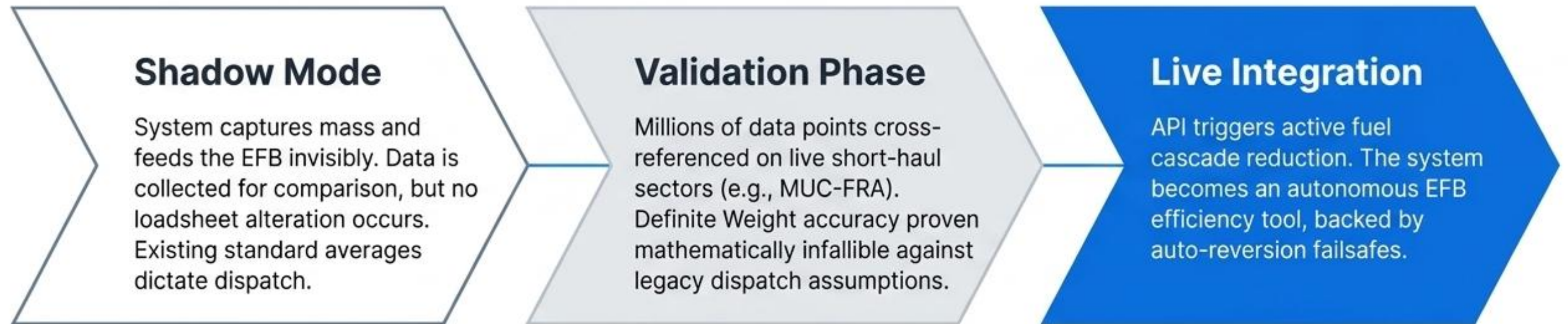
Eliminates Last-Minute Change (LMC) delays. If a checked bag boards but the passenger does not, the gate sensor catches the discrepancy and instantly auto-recalculates the CG in the pilot's EFB, preventing costly manual ground delays.

A Sustainability-as-a-Service model removes upfront capital expenditure and aligns our success with airline savings



We do not ask airlines for millions in upfront hardware costs. We provide the sensor ecosystem for free, monetizing strictly through the verified operational efficiency we generate.

Implementation prioritizes safety: launching as a shadow decision-support layer before activating fuel reduction



The system must fail safe, not fail live. Definite Weight begins as an invisible safety net, earning operational trust before reducing a single drop of fuel.

A uniquely cross-disciplinary technical background specifically engineered to execute the EASA fuel cascade



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Operating as Solo Founder

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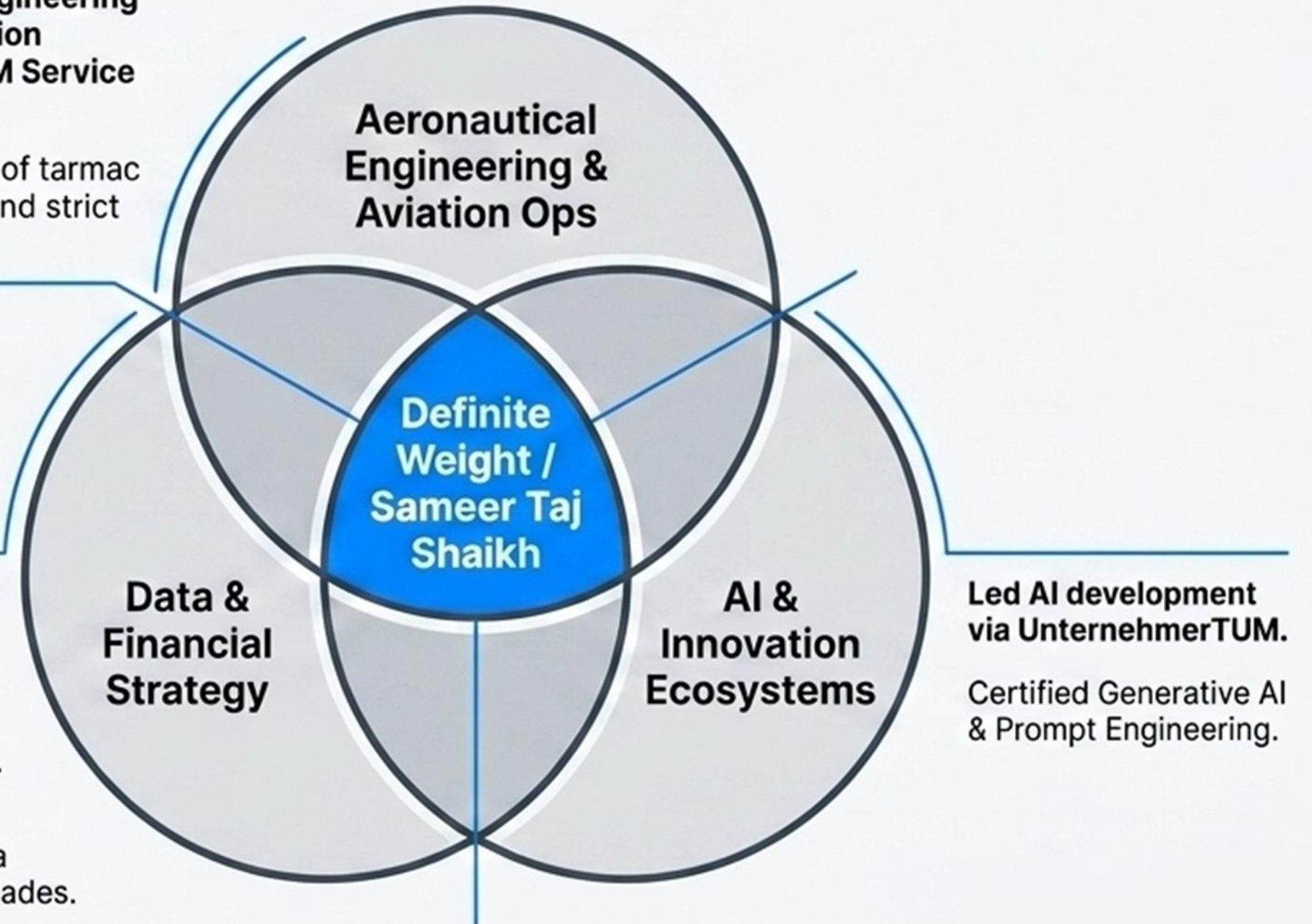
B.E. Aeronautical Engineering (Distinction) & Aviation Management at M+M Service GmbH (Munich).

Deep understanding of tarmac bottleneck realities and strict aviation physics.

MSc Management Candidate at TUM.

Expertise in BI, financial forecasting, Power BI/SQL/Python.

Equipped to build complex financial data models for EASA cascades.



The transition to precision-based flight operations is a financial and environmental imperative for the CORSIA 2026 era

Global carbon offset mandates are tightening. Airlines can no longer afford the fuel-to-carry-fuel penalty.

By eliminating the Uncertainty Tax, Definite Weight instantly transforms regulatory compliance from a massive tax liability into a measurable sustainability win.



The airline that knows its Definite Weight is the only airline equipped to survive the decarbonization mandates of the next decade.



Sameer Taj Shaikh | Solo Founder | Definite Weight™